LAPR4
DEVELOPMENT PROCESS

SPRINT PLANNING

- In the first LAPR4 class of the sprint, the group shall plan the activities for sprint
- There should be written evidence of
  - The list of user stories to be implemented in the sprint.
  - The tasks needed to implement the user stories. Each task will have a corresponding issue created in the repository.
  - The deadline for each task (week of the sprint) and the element responsible for its implementation.

WORKING

- All the team's work must be reflected in the git repository
  - Planning
  - Issues
  - Design artifacts
  - Source code (code and tests)
  - Commits
- All commits must be related to issues (address or close an issue)
- Every team member is expected to have several commits each week
  - In a continuous integration environment, team members are expected to commit daily

WEEKLY SCRUM

- On LAPR4 classes 2 to 4 of the sprint the team will have a weekly scrum meeting
- There should be written evidence of
  - The group's assessment of the fulfilment of the sprint's plan
  - Replanning of sprint's tasks
- The LAPR4 teacher acts as the teams' Scrum Master and will provide feedback to the team about its performance

SPRINT REVIEW AND RETROSPECTIVE

- The sprint review and retrospective will take place in the first LAPR4 class after the sprint's end
- The team will present the sprint's results to the LAPR4 teacher
  - 10 min presentation (maximum)
  - Focus on the development process and teamwork
  - Quality of the implementation of the user stories will be assessed in the UCs related to the user story
- It will be followed by the next sprint planning ceremony

CONTINUOUS INTEGRATION (I)

- The team must use the provided repository in GitHub
- Only one branch is used, as this is a new application
- Every commit is done to the main branch
  - The team member is totally responsible for the commits it make
  - Before pushing commits to the repository, the team member must pull changes from the main repository and compile and run the tests.
  - Any commit must pass the compilation and tests.

    Failure to do so is regarded as a major misbehave. If this happens more than once, the student will be excluded from the project.

CONTINUOUS INTEGRATION (II)

- The Java project is in Maven, and the team must configure a Linux-based continuous integration server in GitHub
- The test coverage (unit tests in JUnit 5) of controllers and domain must be above 90%.
- Use JUnit best practices and use appropriate frameworks for test isolation (e.g., Mokito).
- Tests should follow standard test patterns, like the well established Arrange-Act-Assert (AAA) pattern.
- Design the tests before developing the code, so that they are aligned with business requirements.